

# Equations and Inequalities

Solutions



Curriculum Ready

ACMNA: 234, 235, 236, 237, 240



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## Page 4 questions

1. Choose whether the following equations are linear or quadratic by circling the appropriate word (don't solve):

a  $x + 3 = 6$  is a linear / ~~quadratic~~ equation

(the highest power of  $x$  is 1 so this is a linear equation)

b  $3x^2 = 12$  is a ~~linear~~ / quadratic equation

(the highest power of  $x$  is not 1 (it's 2) so this is not a linear equation)

c  $3x = 12$  is a linear / ~~quadratic~~ equation

(the highest power of  $x$  is 1 so this is a linear equation)

d  $\frac{5x}{2} = 3$  is a linear / ~~quadratic~~ equation

(the highest power of  $x$  is 1 so this is a linear equation)

2. Solve these linear equations:

a  $x + 4 = 7$

$x + 4 - 4 = x + 7 - 4$  ← Subtract 4 from both sides

$x = 3$

b  $x + 3 = 10$

$x + 3 - 3 = 10 - 3$  ← Subtract 3 from both sides

$x = 7$

c  $x - 2 = 13$

$x - 2 + 2 = 13 + 2$  ← Add 2 to both sides

$x = 15$

d  $x - 6 = 10$

$x - 6 + 6 = 10 + 6$  ← Add 6 to both sides

$x = 16$

## Page 5 questions

3. Find the value of the variable in each of these linear equations:

a  $2x = 8$

$$\frac{2x}{2} = \frac{8}{2}$$

Divide both sides by 2

$$x = 4$$

b  $5x = 35$

$$\frac{5x}{5} = \frac{35}{5}$$

Divide both sides by 5

$$x = 7$$

c  $\frac{x}{4} = 5$

$$\frac{x}{4} \times 4 = 5 \times 4$$

Multiply both sides by 4

$$x = 20$$

d  $\frac{x}{3} = 8$

$$\frac{x}{3} \times 3 = 8 \times 3$$

Multiply both sides by 3

$$x = 24$$

e  $\frac{3x}{2} = 9$

$$\frac{3x}{2} \times 2 = 9 \times 2$$

Multiply both sides by 2

$$3x = 18$$

$$\frac{3x}{3} = \frac{18}{3}$$

Divide both sides by 3

$$x = 6$$

f  $\frac{5x}{3} = 5$

$$\frac{5x}{3} \times 3 = 5 \times 3$$

Multiply both sides by 3

$$5x = 15$$

$$\frac{5x}{5} = \frac{15}{5}$$

Divide both sides by 5

$$x = 3$$

## Page 6 questions

4. Solve for the variable in each linear equation:

a  $4x + 3 = 23$

$4x - 3 = 23 - 3$  ← Subtract 3 from both sides

$4x = 20$

$\frac{4x}{4} = \frac{20}{4}$  ← Divide both sides by 4

$x = 5$

b  $2a - 5 = 9$

$2a - 5 + 5 = 9 + 5$  ← Add 5 to both sides

$2a = 14$

$\frac{2a}{2} = \frac{14}{2}$  ← Divide both sides by 2

$a = 7$

c  $5m + 6 = 31$

$5m + 6 - 6 = 31 - 6$  ← Subtract 6 from both sides

$5m = 25$

$\frac{5m}{5} = \frac{25}{5}$  ← Divide both sides by 5

$m = 5$

d  $-4 + 7n = 24$

$-4 + 4 + 7n = 24 + 4$  ← Add 4 to both sides

$7n = 28$

$\frac{7n}{7} = \frac{28}{7}$  ← Divide both sides by 7

$n = 4$

e  $-2k = 10$

$-\frac{2k}{-2} = \frac{10}{-2}$  ← Divide both sides by -2

$k = -5$

f  $-m + 4 = 6$  (Hint:  $-m = -1m$ )

$-m + 4 - 4 = 6 - 4$  ← Subtract 4 from both sides

$-m = 2$

$\frac{-1m}{-1} = \frac{2}{-1}$  ← Divide both sides by -1

$m = -2$

## Page 7 questions

5. Solve for the variable in each linear equation:

a  $\frac{y}{4} = 4$

$\frac{y}{4} \times 4 = 4 \times 4$  ← Multiply both sides by 4

$y = 16$

b  $\frac{5x}{4} = 15$

$\frac{5x}{4} \times 4 = 15 \times 4$  ← Multiply both sides by 4

$5x = 60$

$\frac{5x}{5} = \frac{60}{5}$  ← Divide both sides by 5

$x = 12$

c  $\frac{2p}{9} = 6$

$\frac{2p}{9} \times 9 = 6 \times 9$  ← Multiply both sides by 9

$2p = 54$

$\frac{2p}{2} = \frac{54}{2}$  ← Divide both sides by 2

$p = 27$

d  $\frac{4d}{11} = 8$

$\frac{4d}{11} \times 11 = 8 \times 11$  ← Multiply both sides by 11

$4d = 88$

$\frac{4d}{4} = \frac{88}{4}$  ← Divide both sides by 4

$d = 22$

e  $\frac{7x}{3} = -14$

$\frac{7x}{3} \times 3 = -14 \times 3$  ← Multiply both sides by 3

$7x = -42$

$\frac{7x}{7} = \frac{-42}{7}$  ← Divide both sides by 7

$x = -6$

f  $\frac{10m}{4} = -25$

$\frac{10m}{4} \times 4 = -25 \times 4$  ← Multiply both sides by 4

$10m = -100$

$\frac{10m}{10} = \frac{-100}{10}$  ← Divide both sides by 10

$m = -10$

6. Solve these quadratic equations:

a  $y^2 - 9 = 0$

$y^2 - 9 + 9 = 0 + 9$  ← Add 9 to both sides

$y^2 = 9$  ← Since  $-3^2 = 9$  and  $3^2 = 9$

$y^2 = (\pm 3)^2$  ← Find the square root of both sides

$y = \pm 3$

so  $y = -3$  or  $y = 3$

b  $4x^2 = 100$

$\frac{4x^2}{4} = \frac{100}{4}$  ← Divide both sides by 4

$x^2 = 25$

$x^2 = (\pm 5)^2$  ← Find the square root of both sides

$x = \pm 5$

so  $x = 5$  or  $x = -5$

## Page 11 questions

1. In these linear equations the variable appears on both sides. Solve for the missing value:

a

$$2u - 10 = 3u$$

$$2u - 3u - 10 = 3u - 3u \quad \leftarrow \text{Subtract } 3u \text{ from both sides}$$

$$-1u - 10 + 10 = 0 + 10 \quad \leftarrow \text{Add 10 to both sides}$$

$$\frac{-1u}{-1} = \frac{10}{-1} \quad \leftarrow \text{Divide both sides by } -1$$

$$u = -10$$

b

$$7x - 18 = 3x + 10$$

$$7x - 3x - 18 = 3x - 3x + 10 \quad \leftarrow \text{Subtract } 3x \text{ from both sides}$$

$$4x - 18 + 18 = 10 + 18 \quad \leftarrow \text{Add 18 to both sides}$$

$$4x = 28$$

$$\frac{4x}{4} = \frac{28}{4} \quad \leftarrow \text{Divide both sides by 4}$$

$$x = 7$$

c

$$3(x + 2) = -3$$

$$3x + 6 = -3 \quad \leftarrow \text{Expand the brackets}$$

$$3x + 6 - 6 = -3 - 6 \quad \leftarrow \text{Subtract 6 from both sides}$$

$$3x = -9$$

$$\frac{3x}{3} = \frac{-9}{3} \quad \leftarrow \text{Divide both sides by 3}$$

$$x = -3$$

d

$$4y + 18 = 12 - 2y$$

$$4y + 2y + 18 = 12 - 2y + 2y \quad \leftarrow \text{Add } 2y \text{ to both sides}$$

$$6y + 18 - 18 = 12 - 18 \quad \leftarrow \text{Subtract 18 from both sides}$$

$$6y = -6 \quad \leftarrow \text{Divide both sides by 6}$$

$$y = -1$$

e

$$10(n - 6) = 2(10 + n)$$

$$10n - 60 = 20 + 2n \quad \leftarrow \text{Expand the brackets}$$

$$10n - 2n - 60 = 20 + 2n - 2n \quad \leftarrow \text{Subtract } 2n \text{ from both sides}$$

$$8n - 60 = 20$$

$$8n - 60 + 60 = 20 + 60 \quad \leftarrow \text{Add 60 to both sides}$$

$$8n = 80$$

$$\frac{8n}{8} = \frac{80}{8} \quad \leftarrow \text{Divide both sides by 8}$$

$$n = 10$$

## Page 11 questions

1. In these linear equations the variable appears on both sides. Solve for the missing value:

f  $6m - 4 = -5(m + 3)$  ← Expand the brackets  
 $6m - 4 = -5m - 15$   
 $6m + 5m - 4 = -5m + 5m - 15$  ← Add  $5m$  to both sides  
 $11m - 4 = -15$   
 $11m - 4 + 4 = -15 + 4$  ← Add 4 to both sides  
 $11m = -11$   
 $\frac{11m}{11} = \frac{-11}{11}$  ← Divide both sides by 11  
 $m = -1$

g  $8(k - 4) - 5k + 3 = 4$   
 $8k - 32 - 5k + 3 = 4$  ← Expand the brackets  
 $3k - 29 = 4$  ← Collect like terms  
 $3k - 29 + 29 = 4 + 29$  ← Add 29 to both sides  
 $3k = 33$   
 $\frac{3k}{3} = \frac{33}{3}$  ← Divide both sides by 3  
 $k = 11$

h  $5(2y - 1) - 6(y - 2) + 3 = 6$  ← Expand the brackets  
 $10y - 5 - 6y + 12 + 3 = 6$  ← Collect like terms  
 $4y + 10 = 6$   
 $4y + 10 - 10 = 6 - 10$  ← Subtract 10 from both sides  
 $4y = -4$   
 $\frac{4y}{4} = \frac{-4}{4}$  ← Divide both sides by 4  
 $y = -1$



## Page 11 questions

1. In these linear equations the variable appears on both sides. Solve for the missing value:

i  $8t - (2t - 18) = -12$

$$8t - 2t + 18 = -12$$

Expand the brackets

$$6t + 18 = -12$$

$$6t + 18 - 18 = -12 - 18$$

Subtract 18 from both sides

$$6t = -30$$

$$\frac{6t}{6} = \frac{-30}{6}$$

Divide both sides by 6

$$t = -5$$

j  $2(a + 3) - 3(a + 4) = -10$

$$2a + 6 - 3a - 12 = -10$$

Expand the brackets

$$-a - 6 = -10$$

Collect like terms

$$-a - 6 + 6 = -10 + 6$$

Add 6 to both sides

$$-a = -4$$

$$\frac{-a}{-1} = \frac{-4}{-1}$$

Divide both sides by  $-1$

$$a = 4$$

2. Find Ivan's mistake when he tried to solve this equation?

$$3(h + 2) = 2(h + 1) + 5$$

$$3h + 2 = 2h + 2 + 5$$

$$3h + 2 = 2h + 7$$

Oops - Expanded the brackets incorrectly

should be  $3h + 6 = 2h + 2 + 5$

Correct answer then comes out to  $h = 1$

$$3h + 2 - 2h - 2 = 2h + 7 - 2h - 2$$

$$h = 5$$

## Page 12 questions

3. Solve these linear equations which contain fractions:

a  $\frac{x}{8} - 1 = 4$

$$\frac{x}{8} - 1 + 1 = 4 + 1 \quad \leftarrow \text{Add 1 to both sides}$$

$$\frac{x}{8} = 5$$

$$\frac{x}{8} \times 8 = 5 \times 8 \quad \leftarrow \text{Multiply both sides by 8}$$

$$x = 40$$

b  $\frac{n}{2} + 4 = \frac{5n}{12} + 5 \quad \leftarrow \text{Lowerst common Denominator (LCD) is 12}$

$$12 \times \left( \frac{n}{2} + 4 \right) = 12 \times \left( \frac{5n}{12} + 5 \right) \quad \leftarrow \text{Multiply both sides by (LCD)}$$

$$\frac{12n}{2} + 48 = \frac{60n}{12} + 60 \quad \leftarrow \text{Expand brackets}$$

$$6n + 48 = 5n + 60$$

$$6n - 5n + 48 - 48 = 5n - 5n + 60 - 48$$

$$n = 12$$

c  $\frac{3b+4}{5} = -4$

$$3b + 4 = 5 \times -4 \quad \leftarrow \text{Multiply both sides by denominator 5}$$

$$3b + 4 = -20$$

$$3b + 4 - 4 = -20 - 4$$

$$3b = -24$$

$$b = -8$$

## Page 12 questions

3. Solve these linear equations which contain fractions:

d  $c + \frac{c}{2} = 12$

$$2 \times \left(c + \frac{c}{2}\right) = 2 \times 12 \quad \leftarrow \text{Multiply both sides by LCD (2)}$$

$$2c + c = 24$$

$$3c = 24$$

$$\frac{3c}{3} = \frac{24}{3} \quad \leftarrow \text{Divide both sides by 3}$$

$$c = 8$$

e  $\frac{16r + 2}{5} = 10$

$$16r + 2 = 10 \times 5 \quad \leftarrow \text{Multiply both sides by 5}$$

$$16r + 2 = 50$$

$$16r + 2 - 2 = 50 - 2$$

$$16r = 48$$

$$\frac{16r}{16} = \frac{48}{16} \quad \leftarrow \text{Divide both sides by 16}$$

$$r = 3$$

f  $\frac{m}{2} + \frac{m}{3} = 5$

$$6 \times \left(\frac{m}{2} + \frac{m}{3}\right) = 6 \times 5 \quad \leftarrow \text{Multiply both sides by LCD}$$

$$\frac{6m}{2} + \frac{6m}{3} = 30$$

$$3m + 2m = 30$$

$$5m = 30$$

$$\frac{5m}{5} = \frac{30}{5} \quad \leftarrow \text{Divide both sides by 5}$$

$$m = 6$$

## Page 13 questions

4. Solve these linear equations which contain fractions:

a  $\frac{7q+5}{3} = \frac{4q-30}{6}$  ← LCD is 6

$$6 \times \left( \frac{7q+5}{3} \right) = 6 \times \left( \frac{4q-30}{6} \right)$$

← Multiply by LCD

$$2 \times (7q+5) = 4q-30$$

$$14q+10 = 4q-30$$

$$14q-4q+10-10 = 4q-30-10$$

$$10q = -40$$

$$q = -\frac{40}{10}$$

$$q = -4$$

b  $\frac{5u+5}{6} + \frac{5u+10}{9} + 2 = 7$  ← LCD is 18

$$18 \times \left( \frac{5u+5}{6} \right) + 18 \times \left( \frac{5u+10}{9} \right) + 18 \times 2 = 18 \times 7$$

← Multiply by LCD

$$3 \times (5u+5) + 2 \times (5u+10) + 36 = 126$$

$$15u+15+10u+20+36 = 126$$

$$25u+71 = 126$$

$$25u+71-71 = 126-71$$

$$25u = 55$$

$$\frac{25u}{25} = \frac{55}{25}$$

$$u = \frac{11}{5}$$

c  $\frac{2g}{3} + \frac{g}{10} = \frac{1}{2} + \frac{3g}{4}$  ← LCD is 60

$$60 \times \frac{2g}{3} + 60 \times \frac{g}{10} = 60 \times \frac{1}{2} + 60 \times \frac{3g}{4}$$

← Multiply by (LCD)

$$20 \times 2g + 6g = 30 + 15 \times 3g$$

$$40g + 6g = 30 + 45g$$

$$46g - 45g = 30 + 45g - 45g$$

$$g = 30$$

## Page 13 questions

d  $2 = \frac{6}{x}$

$$2 \times x = \frac{6}{x} \times x$$

Multiply both sides by the denominator

$$2x = 6$$

$$\frac{2x}{2} = \frac{6}{2}$$

$$x = 3$$

e  $\frac{6}{8d-2} = 1$

$$(8d-2) \times \frac{6}{8d-2} = 1 \times (8d-2)$$

Multiply both sides by the denominator

$$6 = 8d - 2$$

$$6 + 2 = 8d - 2 + 2$$

You could rewrite this as  $8d - 2 = 6$  if you prefer

$$8 = 8d$$

$$\frac{8}{8} = \frac{8d}{8}$$

$$d = 1$$

f  $\frac{3k-45}{k} = 2$

$$k \times \frac{3k-45}{k} = k \times 2$$

Multiply both sides by the denominator

$$3k - 45 = 2k$$

$$3k - 2k + 45 = 2k - 2k + 45$$

$$k = 45$$

## Page 14 questions

5. Three times a number is 45. What is the number?

Let the number be  $n$

$$3n = 45$$

$$\frac{3n}{3} = \frac{45}{3}$$

$$n = 15$$

6. Claire, Leanne and Lindsay are sisters. Claire is two years older than Leanne and Leanne is 4 years older than Lindsay. The sum of all their ages is 54. How old is each sister?

Let Leanne's age be  $x$

$$\text{Ages: Claire} + \text{Leanne} + \text{Lindsay} = 54$$

$$(x + 2) + x + (x - 4) = 54$$

$$3x - 2 = 54$$

$$3x - 2 + 2 = 54 + 2$$

$$3x = 56$$

$$\frac{3x}{3} = \frac{56}{3}$$

$$x = 18\frac{2}{3}$$

So Leanne is  $18\frac{2}{3}$ , Claire is  $20\frac{2}{3}$  and Lindsay is  $14\frac{2}{3}$

7. Charlie has been collecting stamps which he keeps in two separate books. The second book has 7 more than triple the stamps of the first book. If he has 35 stamps in total (from both books) then:

Let the number of stamps in the first book be  $x$

The number of stamps in the second book is  $3x + 7$

$$\text{Book 1} + \text{book 2} = \text{total stamps}$$

$$x + 3x + 7 = 35$$

$$4x + 7 = 35$$

$$4x + 7 - 7 = 35$$

$$4x + 7 - 7 = 35 - 7$$

$$4x = 28$$

$$\frac{4x}{4} = \frac{28}{4}$$

$$x = 7$$

- a How many stamps are in the first book? 7
- b How many stamps are in the second book?  $35 - 7 = 28$

**Page 14 questions**

8. Victor has a bag filled with  $2c$  and  $5c$  coins. He has 2 more coins worth  $2c$  than the coins worth  $5c$ . How many  $2c$  coins does Victor have if all his coins sum to  $88c$ ?

Let  $n$  be the number of  $5c$  coins. He has  $(n + 2)$  of  $2c$  coins.

They all add up to  $88c$  so,

$$(n + 2) \times 2 + n \times 5 = 88$$

$$2n + 4 + 5n = 88$$

$$7n + 4 = 88$$

$$7n + 4 - 4 = 88 - 4$$

$$7n = 84$$

$$\frac{7n}{7} = \frac{84}{7}$$

$$n = 12$$

So he has 12 coins worth of  $5c$  and 14 coins worth of  $2c$ .

**Page 18 questions**

1. Identify if the following are true or false:

a  $6 > 3$       True

b  $5 < 8$       True

c  $3 > 8$       False

d  $-2 < -5$       False

e  $-4 > 4$       False

f  $-8 < -4$       True

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2. Solve these inequalities:

a 
$$\begin{aligned}x + 3 &< 4 \\x + 3 - 3 &< 4 \\x &< 1\end{aligned}$$

b 
$$\begin{aligned}x - 4 &\geq 5 \\x - 4 + 4 &\geq 5 + 4 \\x &\geq 9\end{aligned}$$

c 
$$\begin{aligned}m + 7 &\geq -4 \\m + 7 - 7 &\geq -4 - 7 \\m &\geq -11\end{aligned}$$

d 
$$\begin{aligned}p - 10 &\leq -8 \\p - 10 + 10 &\leq -8 + 10 \\p &\leq 2\end{aligned}$$



## Page 18 questions

2. Solve these inequalities:

$$\begin{aligned}\text{e} \quad 5q &\leq 35 \\ \frac{5q}{5} &\leq \frac{35}{5} \\ q &\leq 7\end{aligned}$$

$$\begin{aligned}\text{f} \quad 4h + 3 &> 51 \\ 4h + 3 - 3 &> 51 - 3 \\ 4h &> 48 \\ \frac{4h}{4} &> \frac{48}{4} \\ h &> 12\end{aligned}$$

$$\begin{aligned}\text{g} \quad 5x &> 24 + x \\ 5x - x &> 24 + x - x \\ 4x &> 24 \\ \frac{4x}{4} &> \frac{24}{4} \\ x &> 6\end{aligned}$$

$$\begin{aligned}\text{h} \quad 3(x - 1) &\leq 3(1 - x) \\ 3x - 3 &\leq 3 - 3x \\ 3x + 3x - 3 + 3 &\leq 3 + 3 - 3x + 3x \\ 6x &\leq 6 \\ x &\leq 1\end{aligned}$$

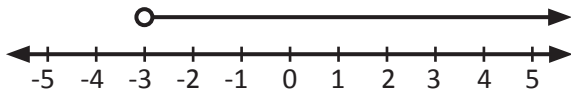
$$\begin{aligned}\text{i} \quad \frac{4h - 8}{2} &< -2 \\ 2 \times \left( \frac{4h - 8}{2} \right) &< -2 \times 2 \\ 4h - 8 &< -4 \\ 4h - 8 + 8 &< -4 + 8 \\ 4h &< 4 \\ \frac{4h}{4} &< \frac{4}{4} \\ h &< 1\end{aligned}$$

$$\begin{aligned}\text{j} \quad \frac{3x}{4} - \frac{2x}{5} &\geq 14 \\ 20 \times \frac{3x}{4} - 20 \times \frac{2x}{5} &\geq 14 \times 20 \\ 5 \times 3x - 4 \times 2x &\geq 280 \\ 15x - 8x &\geq 280 \\ 7x &\geq 280 \\ \frac{7x}{7} &\geq \frac{280}{7} \\ x &\geq 40\end{aligned}$$

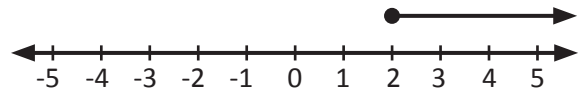
## Page 19 questions

3. Graph these inequalities:

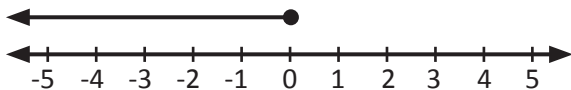
a  $x > -3$



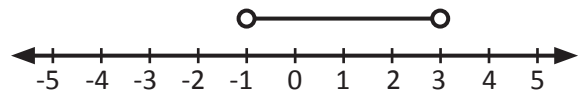
b  $x \geq 2$



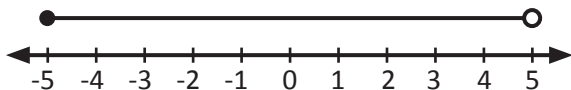
c  $x \leq 0$



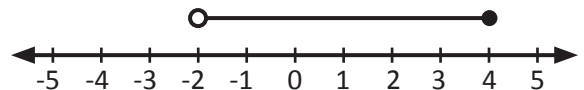
d  $-1 < x < 3$



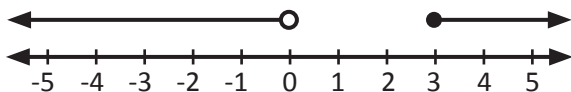
e  $-5 \leq x < 5$



f  $-2 < x \leq 4$



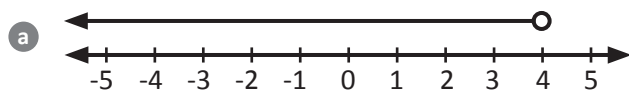
g  $x < 0$  and  $x \geq 3$



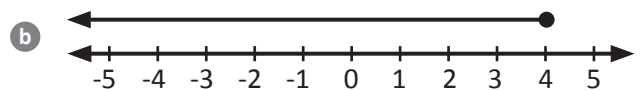
h  $x > 4$  and  $x \leq -1$



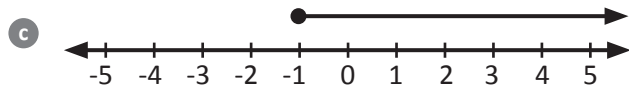
4. Write down the inequality represented by each of the following graphs:



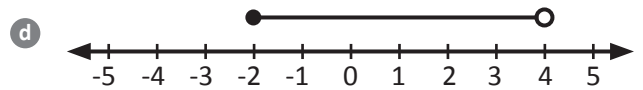
a  $x < 4$



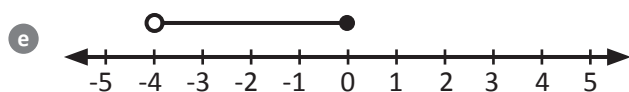
b  $x \leq 4$



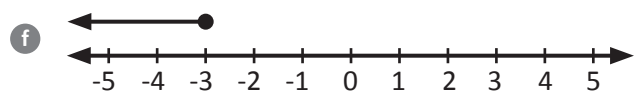
c  $x \geq -1$



d  $-2 \leq x < 4$



e  $-4 < x \leq 0$



f  $x \leq -3$

## Page 20 questions

5. Solve these more complicated linear inequalities, then graph their solution:

a  $5x + 6 > 9(x + 2)$

$$5x + 6 > 9x + 18$$

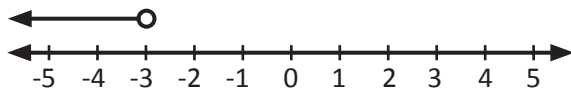
$$5x - 9x + 6 - 6 > 18 - 6$$

$$-4x > 12$$

$$\frac{-4x}{-4} < \frac{12}{-4}$$

$$x < -3$$

Reverse the inequality when dividing by a negative number



b  $7y - 4(y + 4) > 5$

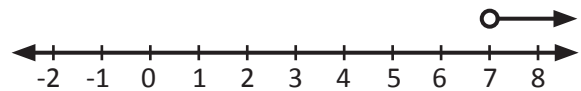
$$7y - 4y - 16 > 5$$

$$3y - 16 + 16 > 5 + 16$$

$$3y > 21$$

$$\frac{3y}{3} > \frac{21}{3}$$

$$y > 7$$



c  $\frac{4 - 3d}{8} \leq 2$

$$8 \times \left( \frac{4 - 3d}{8} \right) \leq 8 \times 2$$

$$4 - 3d \leq 16$$

$$4 - 4 - 3d \leq 16 - 4$$

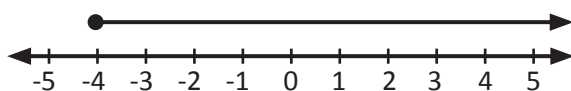
$$-3d \leq 12$$

$$\frac{-3d}{3} \leq \frac{12}{3}$$

$$-d \leq 4$$

$$d \geq -4$$

Reverse the inequality when multiplying by a negative number



## Page 20 questions

5. Solve these more complicated linear inequalities, then graph their solution:

d  $\frac{b}{8} - \frac{b}{3} \leq -5$

$$24 \times \left( \frac{b}{8} - \frac{b}{3} \right) \leq -5 \times 24$$

$$3b - 8b \leq -120$$

$$-5b \leq -120$$

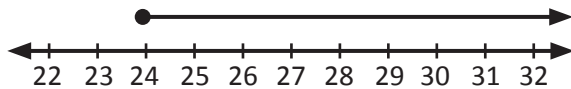
$$-\frac{5b}{5} \leq \frac{120}{5}$$

$$-b \leq -24$$

$$b \geq 24$$

Multiply both sides by LCM of all fractions (24)

Reverse the inequality when multiplying by a negative number.



e  $\frac{a}{-2} \geq \frac{4a+3}{10}$

$$10 \times \left( \frac{a}{-2} \right) \geq 10 \times \left( \frac{4a+3}{10} \right)$$

$$-5a \geq 4a + 3$$

$$-5a - 4a \geq 4a - 4a + 3$$

$$-9a \geq 3$$

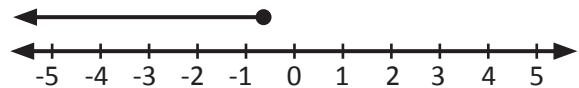
$$-\frac{9a}{9} \geq \frac{3}{9}$$

$$-a \geq \frac{1}{3}$$

$$a \leq -\frac{1}{3}$$

Multiply both sides by LCM of all fractions (10)

Reverse the inequality when multiplying by a negative number



f  $\frac{6c-2}{48} < \frac{1}{2} - \frac{c}{12}$

$$48 \times \left( \frac{6c-2}{48} \right) < 48 \times \frac{1}{2} - 48 \times \frac{c}{12}$$

$$6c - 2 < 24 - 4c$$

$$6c + 4c - 2 + 2 < 24 - 4c + 4c + 2$$

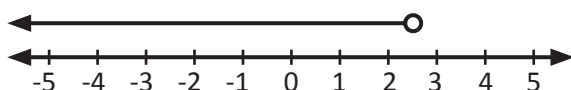
$$10c < 26$$

$$\frac{10c}{10} < \frac{26}{10}$$

$$c < 2.6$$

Multiply both sides by LCM of all fractions (48)

Reverse the inequality when multiplying by a negative number



## Page 26 questions

1. Write down 2 possible solutions for the variables in these equations:

a  $x + y = 4$

Anything where  $x$  and  $y$  add up to 4Eg  $x = 3, y = 1$  or  $x = 2, y = 2$  or  $x = 4, y = 0$  or  $x = 5, y = -1$  or  $x = 3.5, y = 0.5$  etcCan also write these pairs like this:  $(3, 1), (2, 2), (4, 0), (5, -1), (3.5, 0.5)$ 

b  $2a + b = 6$

Anything where twice  $a$ , plus  $b$ , makes 6.Eg  $a = 1, b = 4$ , or  $a = 2, b = 2$  or  $a = 3, b = 0$  or  $a = 4, b = -2$  or  $a = 0.5, b = 5$  etc

c  $3x - 4y = 10$

Anything where 3 times the first number, minus three times the second number, makes 10.

Eg  $x = 6, y = 2$ , or  $x = -2, y = -4$  or  $x = 30, y = 20$  or  $x = 2, y = -1$  or  $x = 1, y = -1.75$  etc

2. Solve for the variables in these simultaneous equations using the substitution method:

a 
$$\begin{array}{l} 2x + y = -1 \quad (1) \\ x - 2y = -4 \quad (2) \end{array}$$

Make one of the variables the subject of (1)

$$y = -1 - 2x$$

Substitute this into (2)

$$x - 2(-1 - 2x) = -4$$

$$x + 2 + 4x = -4$$

$$5x + 2 = -4$$

$$5x = -6$$

$$x = -\frac{6}{5} = -1.2$$

Substitute this  $x$  value into either equation, say (2)

$$x - 2y = -4$$

$$-1.2 - 2y = -4$$

$$-2y = -4 + 1.2$$

$$y = \frac{-2.8}{-2}$$

$$y = 1.4$$

so  $x = -1.2$  and  $y = 1.4$ 

b 
$$\begin{array}{l} 2p + 3q = 10 \quad (1) \\ 2q - 4p = 44 \quad (2) \end{array}$$

Make one of the variables the subject of (2), say  $q$ 

$$2q = 44 + 4p$$

$$\frac{2q}{2} = \frac{44}{2} + \frac{4p}{2}$$

$$q = 22 + 2p$$

Substitute this into (1)

$$2p + 3(22 + 2p) = 10$$

$$2p + 66 + 6p = 10$$

$$8p + 66 - 66 = 10 - 66$$

$$8p = -56$$

$$p = -7$$

Substitute  $p$  back into either equation to solve for  $q$ 

$$2 \times -7 + 3q = 10 \quad (1)$$

$$-14 + 3q = 10$$

$$3q - 14 + 14 = 10 + 14$$

$$3q = 24$$

$$q = 8$$

so  $p = -7$  and  $q = 8$

## Page 27 questions

3. Use the graphical method to solve for these equations:

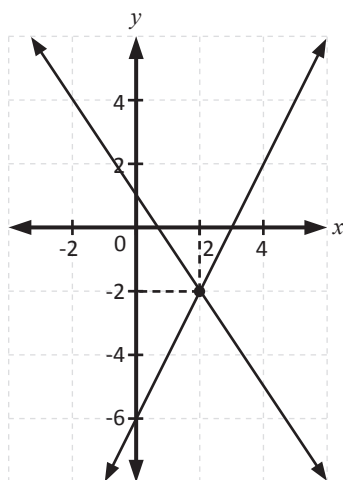
a  $3x + 2y = 2$  ①

$2x - y = 6$  ②

Make  $y$  the subject of both equations

$$y = -\frac{3}{2}x + 1$$

$$y = 2x - 6$$



The lines intersect at  $(2, -2)$  so  $x = 2$  and  $y = -2$

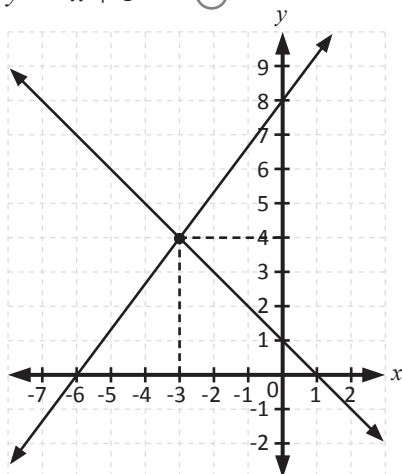
b  $3y - 4x = 24$  ①

$2y + 2x = 2$  ②

Make  $y$  the subject of both equations

$$y = \frac{4}{3}x + 8$$
 ①

$$y = -x + 1$$
 ②



The lines intersect at  $(-3, 4)$  so  $x = -3$  and  $y = 4$

## Page 28 questions

4. Solve for the variables in these simultaneous equations using the elimination method:

$$\begin{aligned} \text{a} \quad 3x - y &= -15 & \textcircled{1} \\ y + 2x &= 0 & \textcircled{2} \end{aligned}$$

In this case we can eliminate  $y$  by adding the two equations together.

$$\begin{aligned} 3x - y &= -15 & \textcircled{1} \\ y + 2x &= 0 & \textcircled{2} \\ 3x + 2x &= -15 & \textcircled{1} + \textcircled{2} \\ 5x &= -15 \\ x &= -3 \end{aligned}$$

Substitute  $x = -3$  into either equation – say  $\textcircled{1}$ .

$$\begin{aligned} 3 \times -3 - y &= -15 \\ -9 - y &= -15 \\ y &= 6 \\ \text{so } x &= -3 \text{ and } y = 6 \end{aligned}$$

$$\begin{aligned} \text{b} \quad b - 4a &= -12 & \textcircled{1} \\ 3a - 2b &= -1 & \textcircled{2} \end{aligned}$$

Make the coefficient of  $b$  the same in both equations; then subtract  $\textcircled{2}$  from  $\textcircled{1}$

$$\begin{aligned} 2b - 8a &= -24 & \textcircled{1} \times 2 \\ 3a - 2b &= -1 & \textcircled{2} \\ \hline -5a &= -25 & (2 \times \textcircled{1}) + \textcircled{2} \\ \frac{-5a}{-5} &= \frac{-25}{-5} \\ a &= 5 \end{aligned}$$

Substitute  $a = 5$  into either equation, say  $\textcircled{2}$

$$\begin{aligned} 3 \times 5 - 2b &= -1 \\ 15 - 2b &= -1 \\ -2b + 15 - 15 &= -1 - 15 \\ -2b &= -16 \\ b &= 8 \end{aligned}$$

So  $a = 5$  and  $b = 8$

## Page 29 questions

5. Solve these simultaneous equations using any method:

$$\text{a } 8c - 8d = 2 \quad (1)$$

$$9c - 8d = 5 \quad (2)$$

-----

$$-c = -3 \quad (1) - (2)$$

$$\therefore c = 3$$

substitute  $c = 3$  into (1) to find  $d$ :

$$\therefore 8(3) - 8d = 2$$

$$\therefore 8d = 22$$

$$\therefore d = \frac{22}{8} = \frac{11}{4}$$

$$\therefore c = 3 \text{ and } d = \frac{11}{4}$$

## Page 30 questions

6. Solve these simultaneous equations using any method:

$$\text{a } 7m = 16 - 6n \quad (1)$$

$$2n = 3m + 16 \quad (2)$$

In (2), make  $n$  the subject (find  $n$  by itself):

$$n = \frac{3m + 16}{2}$$

Substitute this into (1):

$$\therefore 7m = 16 - 6\left(\frac{3m + 16}{2}\right)$$

$$\therefore 7m = 16 - 9m - 48$$

Now, solve for  $m$ :

$$\therefore 16m = -32$$

$$\therefore m = -2$$

Substitute this into (1) to solve for  $n$ :

$$2n = 3(-2) + 16$$

$$n = 5$$



## Page 31 questions

7. Solve these simultaneous equations using any method:

a  $6a - 2b - 10 = 0$  ①

$$2a + 3b - 29 = 0 \quad \text{②}$$

Multiply ② by ① and subtract:

$$6a - 2b - 10 = 0 \quad \text{①}$$

$$6a + 9b - 87 = 0 \quad \text{②}$$

-----

$$-11b = -77$$

$$\therefore b = 7$$

Substitute this into ②:

$$2a + 3(7) - 29 = 0$$

$$2a = 8$$

$$a = 4$$

## Page 32 questions

## 8. Find equations and solve them for these word problems (using any method):

- a The sum of two numbers is 12. The sum of the first number, and double the second number is 16. What are the numbers?

Call the numbers  $a$  and  $b$

$$a + b = 12 \quad \textcircled{1} \text{ (The sum of two numbers is 12)}$$

$$a + 2b = 16 \quad \textcircled{2} \text{ (The sum of the first number, and double the second number is 16)}$$

You can solve this using any of the 3 methods, for example elimination. We can choose to eliminate  $a$  since it already has the same coefficient in both equations.

$$b - 2b = 12 - 16 \quad \textcircled{1} - \textcircled{2}$$

$$-b = -4$$

$$b = 4$$

Substitute  $b = 4$  into  $\textcircled{1}$

$$a + 4 = 12$$

$$a = 8$$

So  $a = 8$  and  $b = 4$

- b Ari is three years older than Eric. In three years from now, Ari will be twice as old as Eric will be. How old are they now?

Lets call Eric's present age,  $e$ , and Ari's present age  $a$

$$a = e + 3 \quad \textcircled{1}$$

$$\frac{a+3}{2} = e+3 \quad \textcircled{2}$$

You can solve this using any of the 3 methods, for example substitution.

$$\frac{e+3+3}{2} = e+3 \quad \text{Substitute } \textcircled{1} \text{ into } \textcircled{2}$$

$$2 \times \left( \frac{e+6}{2} \right) = 2 \times (e+3)$$

$$e+6 = 2e+6$$

$$e = 0$$

Substitute  $e = 0$  into either equation gives  $a = 3$

So Eric is 0; Ari is 3.

## Page 32 questions

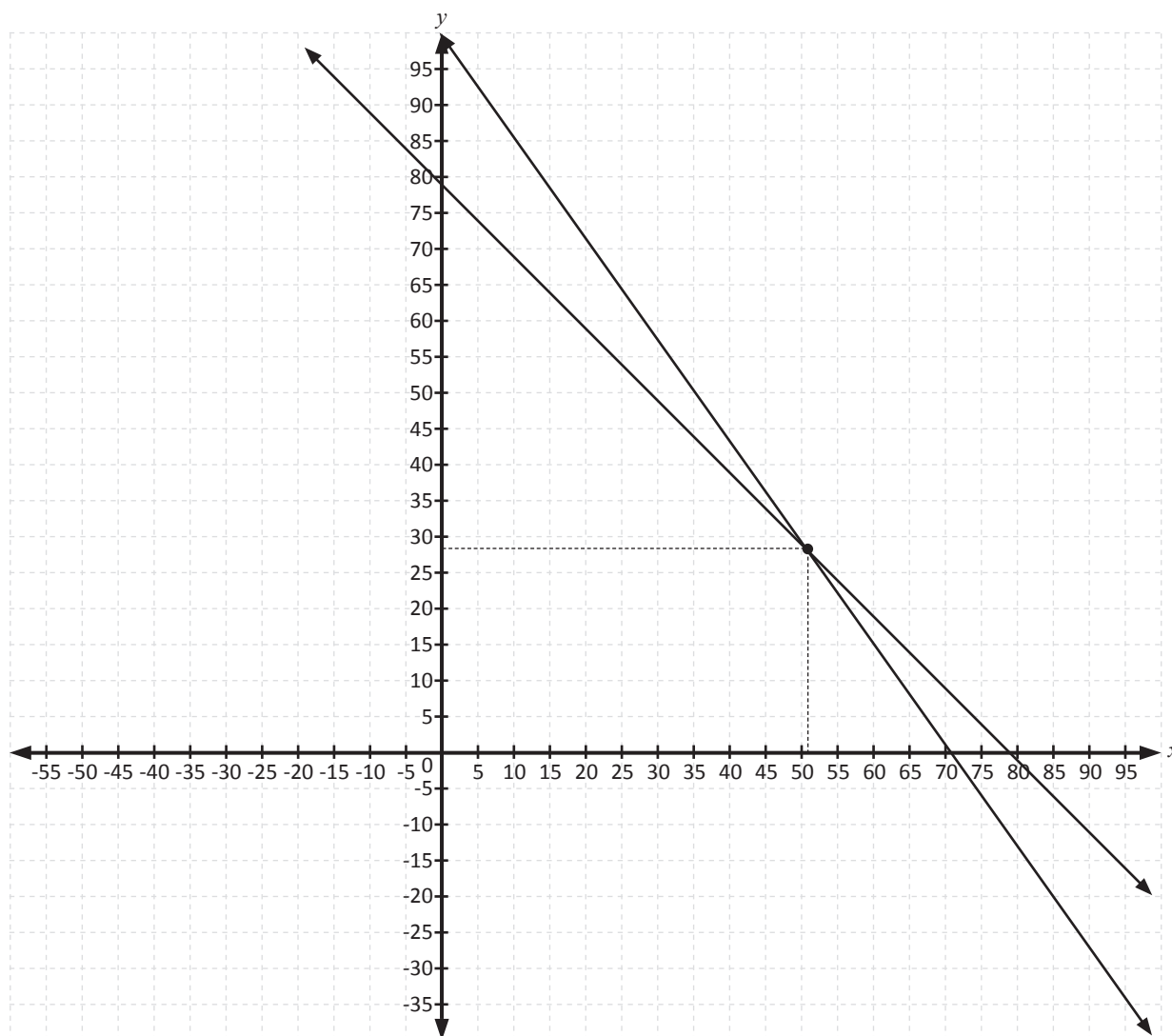
## 8. Find equations and solve them for these word problems (using any method):

- c A restaurant sells two kinds of meals: pizza and pasta. A pizza costs \$14 and a pasta costs \$10. In a single day the restaurant sold 79 meals. If they earned \$994 on this day, how many of each meal was sold?

If we call number of pizzas sold is  $x$  and number of pasta meals sold is  $y$ , then  $x + y = 79$  (79 meals sold in total)

$$14x + 10y = 994 \text{ (\$14 for every pizza + \$10 for every pasta meal = \$994 in total)}$$

You can solve this with any of the 3 methods, for example graphically



Careful measurement will show that  $x = 51$  and  $y = 28$ .

So 51 pizzas and 28 pasta meals were sold.

Note: This method is approximate, and the answer will not always be exact.











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